

NEHA PALAK

neha.palak_ug2023@ashoka.edu.in, 21 years
+91 9709000628 [linkedin.com/in/neha-palak](https://www.linkedin.com/in/neha-palak)



EDUCATION

B.Sc in Computer Science and Mathematics, Minor in Economics	Ashoka University	2027	3.32/4 GPA
Concentration in Business Economics	Hult International Business School	2025	90%
CLASS XII	Loyola High School, Patna	CBSE 2023	82%
CLASS X	Bishop Scott Girls' School, Patna	CBSE 2021	96%

Subjects / Electives: Probability and Statistics, Introduction to Computer Science, Data Structures and Algorithms, Linear Algebra, Microeconomic Theory 1, Foundations, Strategy and Innovation & Impact, Discrete Mathematics, Information Security, Introduction to Machine Learning, Computer Organisation and Systems, Multivariable Calculus, Calculus, Graph Theory and Network Sciences, Computer Vision

Technical Proficiency: Power BI, Git, Pandas, Python, PyTorch, C++, NumPy, SQL Database, MongoDB, Neo4j, MS Excel, MATLAB, LaTeX, SciPy, OpenCV

INTERNSHIPS EXPERIENCE

Program Management Intern | BuildUp Global Apr 2026 – Present

- Mentored **two student teams** on live consulting projects for **Sensio AI**, a healthcare AI and wearable technology startup, overseeing project planning, timelines, and final deliverables.
- Guided the **CS team** in designing a **global tender intelligence system**, including tender discovery and aggregation, keyword-based smart matching logic, global activity mapping across 10–15 countries, and a weekly alert system prototype.
- Guided the **Data Analysis team** in building a **district-level healthcare gap index** for rural India by combining disease prevalence, healthcare infrastructure, and demographic datasets, and developing interactive maps to identify top deployment clusters for wearable interventions.
- Collaborated with the company founder and BuildUp coordinators to ensure project alignment, provided regular feedback to interns, and maintained ownership of all working files and final reports.

Academic Counselor | Young Scholar’s Program – Ashoka University May – Jun, 2025

- Served as **instructor for the TDCS Makerspace Workshop** (2nd edition), teaching **15+ students** topics including cybersecurity, Python fundamentals, encryption, and building advanced mesh networks.
- Supported program operations and academic sessions for **500+ students**.

Media Intern – Office of Admissions | Ashoka University Feb – Aug, 2025

- Managed student engagement initiatives, conducting **150+ office hours** to address queries from incoming undergraduates.
- Ideated, hosted, and presented a **webinar for 400+ incoming students**, providing guidance on academics and campus life.

Student Intern | Koita Centre for Digital Health – Ashoka University Jun – Jul, 2024

- KCDH-A is a research centre dedicated to promoting academic initiatives and conducting research in the field of Digital Health.
- Was one of **10 people selected** for their summer internship program, contributing to research projects in digital health during Summer 2024.
- Worked on **Large Language Models (LLMs)** to enhance understanding and implementation of informed consent in healthcare, showcasing applied machine learning and optimisation concepts.

PROJECTS

Emotion-Conditioned Image Captioning Nov – Dec, 2025

CS-3410: Machine Learning | Instructor: [Prof. Lipika Dey](#)

- Developed a **multimodal image captioning system** generating captions conditioned on both visual content and emotion labels.
- Implemented and compared **CNN–LSTM** and **Vision-Language Transformer (ViT)** architectures.
- Built preprocessing pipelines for **image normalization** and **text tokenization**, vocabulary construction, and padding.
- Experimented with **GloVe**, **FastText**, and **TF-IDF embeddings** and evaluated performance using **BLEU** and **ROUGE metrics**.
- Analyzed loss curves, attention maps, and failure cases to improve caption coherence and emotional grounding.

Child Marriage in India: A Data-Driven Analysis of Socio-Economic Risk Factors April, 2026

CS-3510: Data Science and Management | Instructor: [Prof. Lipika Dey](#) | With Anand Agarwal

- Constructed a **multi-dimensional dataset** of child marriage rates across all Indian states using **15+ Census of India sub-tables** from the 2001 and 2011 census waves.
- Designed a **Currently Married Proportion Rate (CMPR)** index and built a **modular Python data pipeline** to parse, filter, and merge raw census Excel files across SC, ST, total, and religion sub-groups.
- Ran **LASSO regression** across all eight age brackets with **Leave-One-Out Cross-Validation**, using adult brackets as control groups to distinguish child-marriage-specific predictors from general social gradients.
- Performed **hierarchical clustering** (Ward linkage, $k = 3$) to identify high-burden state groups and analysed the West Bengal paradox (high literacy coexisting with persistently elevated CMPR).
- Built and deployed an **interactive web dashboard** (project-bachpan.vercel.app) for non-technical audiences to explore

CMPR patterns across states, age brackets, and sub-groups.

Indian Food Recognition and Detection (Khana Dataset)

April, 2026

CS-4912: Computer Vision | Instructor: *Prof. Raghavendra Singh*

- Fine-tuned a **Vision Transformer (ViT)** model via the **timm** library on a multi-class Indian food dataset, implementing **stratified train/val splits** and **weighted loss** to handle class imbalance.
- Applied **Test-Time Augmentation (TTA)** for more robust inference by averaging predictions across multiple transformed views of each image.
- Integrated **YOLOv8** for object detection on food thali images, filtering detections by COCO food classes and classifying each cropped region using the fine-tuned ViT classifier.
- Implemented a **Bird's Eye View (BEV) pipeline** using multi-strategy corner detection (Canny contours, Hough lines, GrabCut) and **perspective warping** (OpenCV) to normalise overhead tray images before detection.

AWARDS AND ACCOMPLISHMENTS

Dean's List, Spring 2025 | Ashoka University

May 2025

Deans List is an honour awarded when a student's Semester GPA is above 3.65/4.

Commendation Award – Author of Tomorrow | The Wilbur & Niso Smith Foundation

Jul 2023

Apart from the 10 shortlisted entries, 80 were commended among almost 900 participants.

Certificate of Academic Recognition | Government of Bihar

Aug 2021

5 out of 120 students were awarded for excellent performance in CBSE 10th boards examination.

CONFERENCES AND WORKSHOPS

exSPLOre 2025 Workshop | Organized by: Safexpress Centre for Data, Learning and Decision Sciences
Jan 2025

- Selected as **1 of 70 undergraduates** for a five-day workshop on advanced topics in data science, machine learning, and AI.
- Attended tutorials on probability, optimization, and algorithms, and explored research through talks on reinforcement learning, gen AI, and LLMs.
- Engaged in discussions and poster sessions, networking with researchers and peers on quantitative methods.

Metvy Data Analytics Program | Organized by: Metvy Learn

Jan 2024

- Completed a 4-week data analytics program, gaining skills in **Microsoft Excel, SQL, Power BI, and Python**.
- Applied knowledge through assignments testing data analytics concepts and tools.

Dexterity School of Leadership and Entrepreneurship (DexSchool) | Organized by: The Dexterity Global Group
Jun 2023

- Selected as **one of 103 individuals** from a highly competitive pool for an exclusive leadership development program.
- Engaged in a year-long intensive training program focused on honing crucial skills such as communication, research, and public service.
- Exhibited leadership by solving complex problems and building character and temperament for future leadership.